CLASS 8 · CBSE · GANITA PRAKASH PART 2 · FORMULA BANK

Chapter 2: The Baudhāyana–Pythagoras Theorem

Every formula and result of the chapter on one sheet

1	Square on the diagonal	Double the area of the square
2	Isosceles right triangle	$c^2 = 2a^2$, $c = a\sqrt{2}$
3	Root 2	$\sqrt{2} = 1.41421356\ldots$, between 1.414 and 1.415
4	Baudhāyana–Pythagoras theorem	$a^2 + b^2 = c^2$
5	Hypotenuse	$c = \sqrt{a^2 + b^2}$
6	Shorter side	$a = \sqrt{c^2 - b^2}$
7	Baudhāyana triple	Positive integers with $a^2 + b^2 = c^2$
8	Scaling	(ka, kb, kc) is a triple if (a, b, c) is
9	Triples from odd squares	$(n-1)^2 + (2n-1) = n^2$
10	Common triples	$(3,4,5)$, $(5,12,13)$, $(8,15,17)$, $(7,24,25)$
11	Diagonal of a rectangle	$\sqrt{l^2 + b^2}$
12	Diagonal of a square	$a\sqrt{2}$
13	Height of an equilateral triangle	$\frac{\sqrt{3}}{2}s$
14	Area of an equilateral triangle	$\frac{\sqrt{3}}{4}s^2$
15	Fermat's Last Theorem	$x^n + y^n = z^n$ has no natural-number solution for $n > 2$